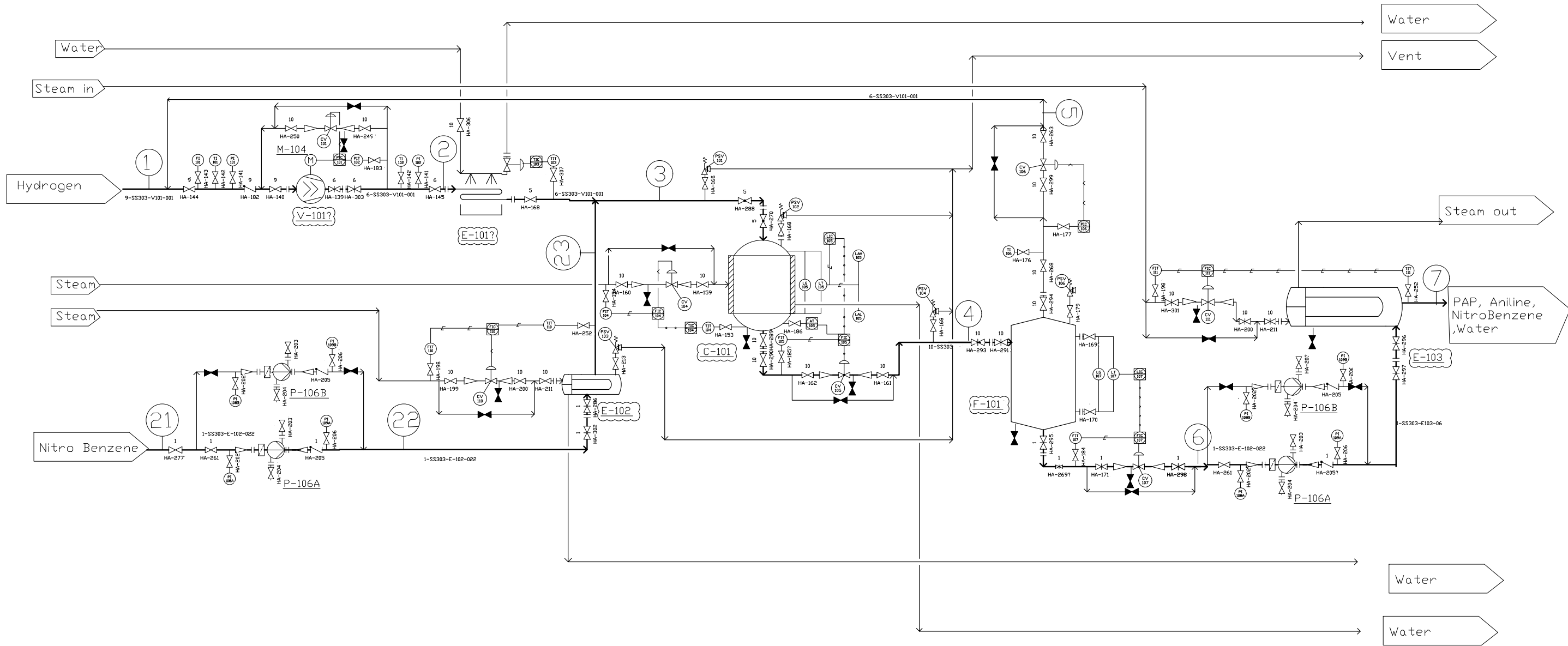




DESCRIPTION OF DIFFERENT STREAM LINES OF PIPING AND INSTRUMENTATION DIAGRAM

Stream Name		1	2	3	4	5	6	7	21	22	23
Properties	Units										
Temperature	C	25	226.0968167	83.9175473	85	25	25	120	25	25.36335903	85
Pressure	bar	1	4	4	4	1	1	1	1	4	4
Mass Flows	kg/hr	201.588	201.588	6357.15	6357.15	96.12611259	6261.023887	6261.023887	6155.562	6155.562	6155.562
HYDROGEN	kg/hr	201.588	201.588	201.588	62.49228	62.48528048	0.006999516	0.006999516	0	0	0
NITRO-BE	kg/hr	0	0	6155.562	2462.2248	0.406641918	2461.818158	2461.818158	6155.562	6155.562	6155.562
PAP	kg/hr	0	0	0	2291.68212	0.001621991	2291.680498	2291.680498	0	0	0
ANILINE	kg/hr	0	0	0	838.15488	0.401891479	837.7529885	837.7529885	0	0	0
ACET-ANH	kg/hr	0	0	0	0	0	0	0	0	0	0
ACETIC	kg/hr	0	0	0	0	0	0	0	0	0	0
CETAMOL	kg/hr	0	0	0	0	0	0	0	0	0	0
WATER	kg/hr	0	0	0	702.59592	32.83067672	669.7652433	669.7652433	0	0	0
SOLID	kg/hr	0	0	0	0	0	0	0	0	0	0
ETHANOL	kg/hr	0	0	0	0	0	0	0	0	0	0
Mass Density	gm/cc	8.12818E-05	0.000194003	0.00847113	0.022678404	0.000118089	1.158381828	0.007311989	1.198493352	1.198140729	1.138592471
Enthalpy Flow	cal/sec	3.657343868	38923.58486	80138.4913	-843155.89	-29199.444	-877560.892	-738081.92	30022.53078	30345.88726	68594.25149
Volume Flow	l/min	41335.18251	17318.25706	12507.4823	4671.955688	13566.93798	90.08290324	14271.1369	85.60139265	85.62658582	90.10484664
Mole Flows	kmol/hr	100	100	150	120	32.82654066	87.17345934	87.17345934	50	50	50

Tag	Type	Description
V-101	V	SCREW COMPRESSOR
M-104	M	ELECTRIC MOTOR
E-101	E	SPRAY COOLER
C-101	C	IMPELLER AGITATOR
F-101	F	CONICAL HEADS VESSEL
E-102	E	SHELL AND TUBE BUNDLE HEAT EXCHANGER WITH U-TUBE
E-103	E	SHELL AND TUBE BUNDLE HEAT EXCHANGER WITH U-TUBE
P-106A	P	CENTRIFUGAL PUMP
P-106B	P	CENTRIFUGAL PUMP

Manufacture of Paracetamol From Nitro Benzene				
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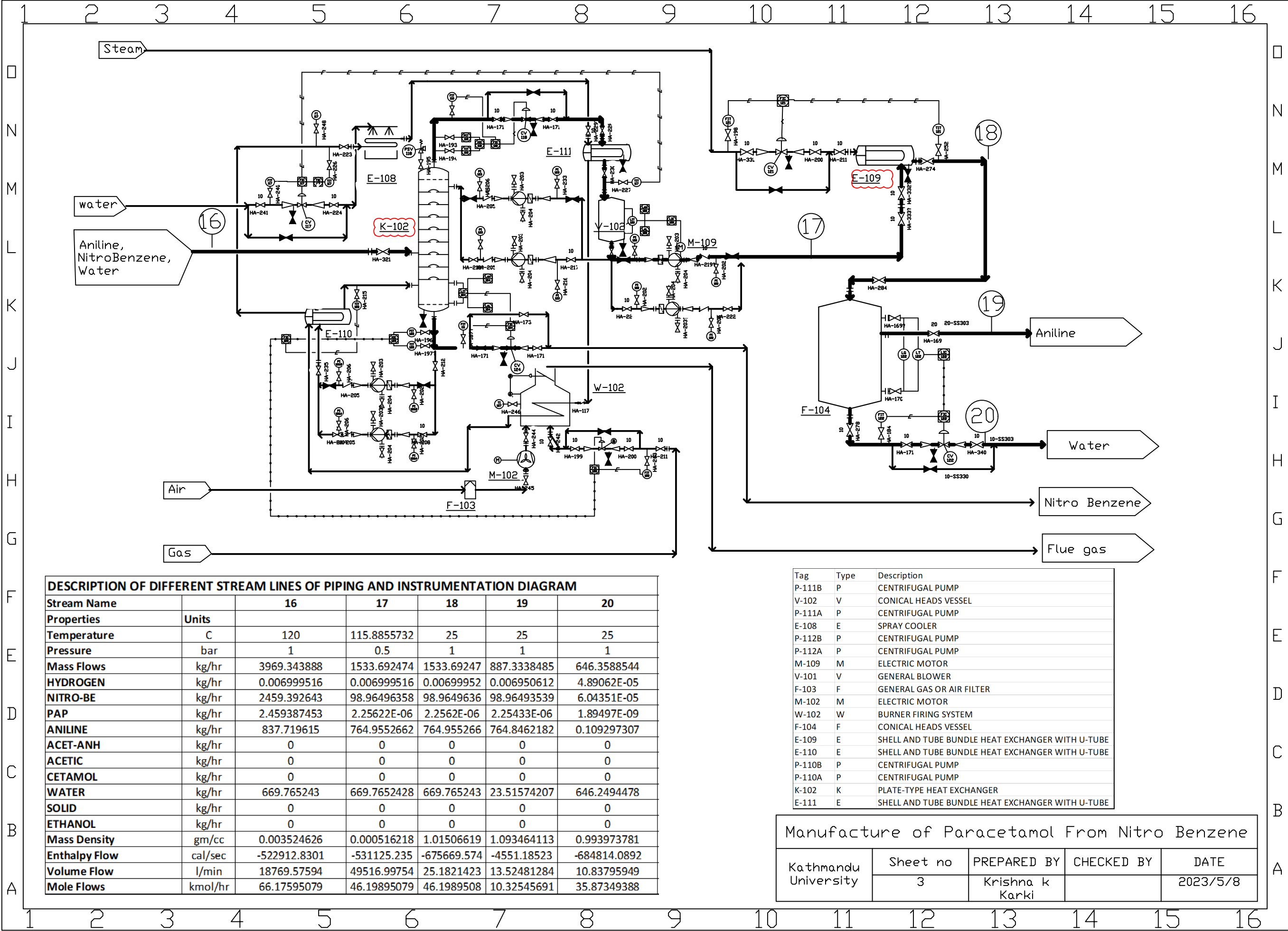
DESCRIPTION OF DIFFERENT STREAM LINES OF PIPING AND INSTRUMENTATION DIAGRAM

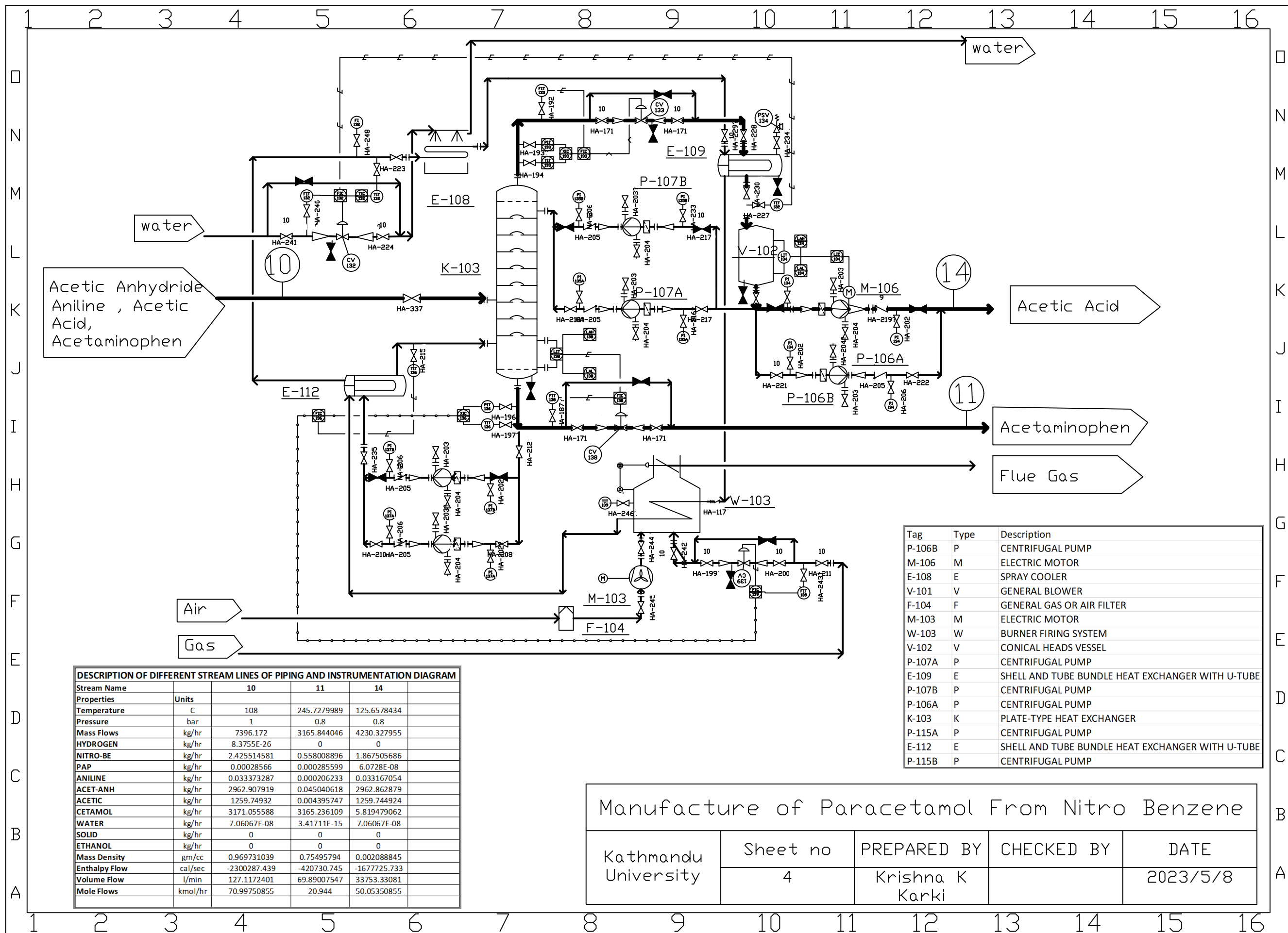
Stream Name		7	8	9	10	12	15	16
Properties	Units							
Temperature	C	120	271.9480585	108	108	25	167.7871749	120
Pressure	bar	1	0.8	1	1	1	0.8	1
Mass Flows	kg/hr	6261.023887	2291.68	7396.172	7396.172	5104.492	3969.343888	3969.343888
HYDROGEN	kg/hr	0.006999516	8.3755E-26	8.3755E-26	8.3755E-26	0	0.006999516	0.006999516
NITRO-BE	kg/hr	2461.818158	2.425514581	2.42551458	2.425514581	0	2459.392643	2459.392643
PAP	kg/hr	2291.680498	2289.221112	2289.22111	0.00028566	0	2.459387453	2.459387453
ANILINE	kg/hr	837.7529885	0.033373287	0.03337329	0.033373287	0	837.719615	837.719615
ACET-ANH	kg/hr	0	0	5104.492	2962.907919	5104.492	0	0
ACETIC	kg/hr	0	0	0	1259.74932	0	0	0
CETAMOL	kg/hr	0	0	0	3171.055588	0	0	0
WATER	kg/hr	669.7652433	7.06067E-08	7.0607E-08	7.06067E-08	0	669.765243	669.765243
SOLID	kg/hr	0	0	0	0	0	0	0
ETHANOL	kg/hr	0	0	0	0	0	0	0
Mass Density	gm/cc	0.007311989	0.932667265	1.01180007	0.969731039	1.074078949	0.001324808	0.003524626
Enthalpy Flow	cal/sec	-738081.9202	-143717.098	-2211998.91	-2300287.44	-2064486.72	-404450.215	-522912.83
Volume Flow	l/min	14271.1369	40.9520824	121.831908	127.1172401	79.20727499	49936.10639	18769.57594
Mole Flows	kmol/hr	87.17345934	20.99750855	70.9975085	70.99750855	50	66.17595079	66.17595079

Tag	Type	Description
P-103B	P	CENTRIFUGAL PUMP
E-105	E	SHELL AND TUBE BUNDLE HEAT EXCHANGER WITH U-TUBE
F-101	F	CONICAL HEADS VESSEL
P-101B	P	CENTRIFUGAL PUMP
P-101A	P	CENTRIFUGAL PUMP
K-101	K	PLATE-TYPE HEAT EXCHANGER
E-106	E	SHELL AND TUBE BUNDLE HEAT EXCHANGER WITH U-TUBE
P-103A	P	CENTRIFUGAL PUMP
P-102B	P	CENTRIFUGAL PUMP
M-105	M	ELECTRIC MOTOR
P-102A	P	CENTRIFUGAL PUMP
E-104	E	SPRAY COOLER
V-101	V	GENERAL BLOWER
F-102	F	GENERAL GAS OR AIR FILTER
V-101	V	ELECTRIC MOTOR
W-101	W	BURNER FIRING SYSTEM
C-102	C	IMPELLER AGITATOR
E-107	E	SHELL AND TUBE BUNDLE HEAT EXCHANGER WITH U-TUBE
P-104B	P	CENTRIFUGAL PUMP
P-104A	P	CENTRIFUGAL PUMP
E-108	E	SHELL AND TUBE BUNDLE HEAT EXCHANGER WITH U-TUBE

Manufacture of Paracetamol From Nitro Benzene

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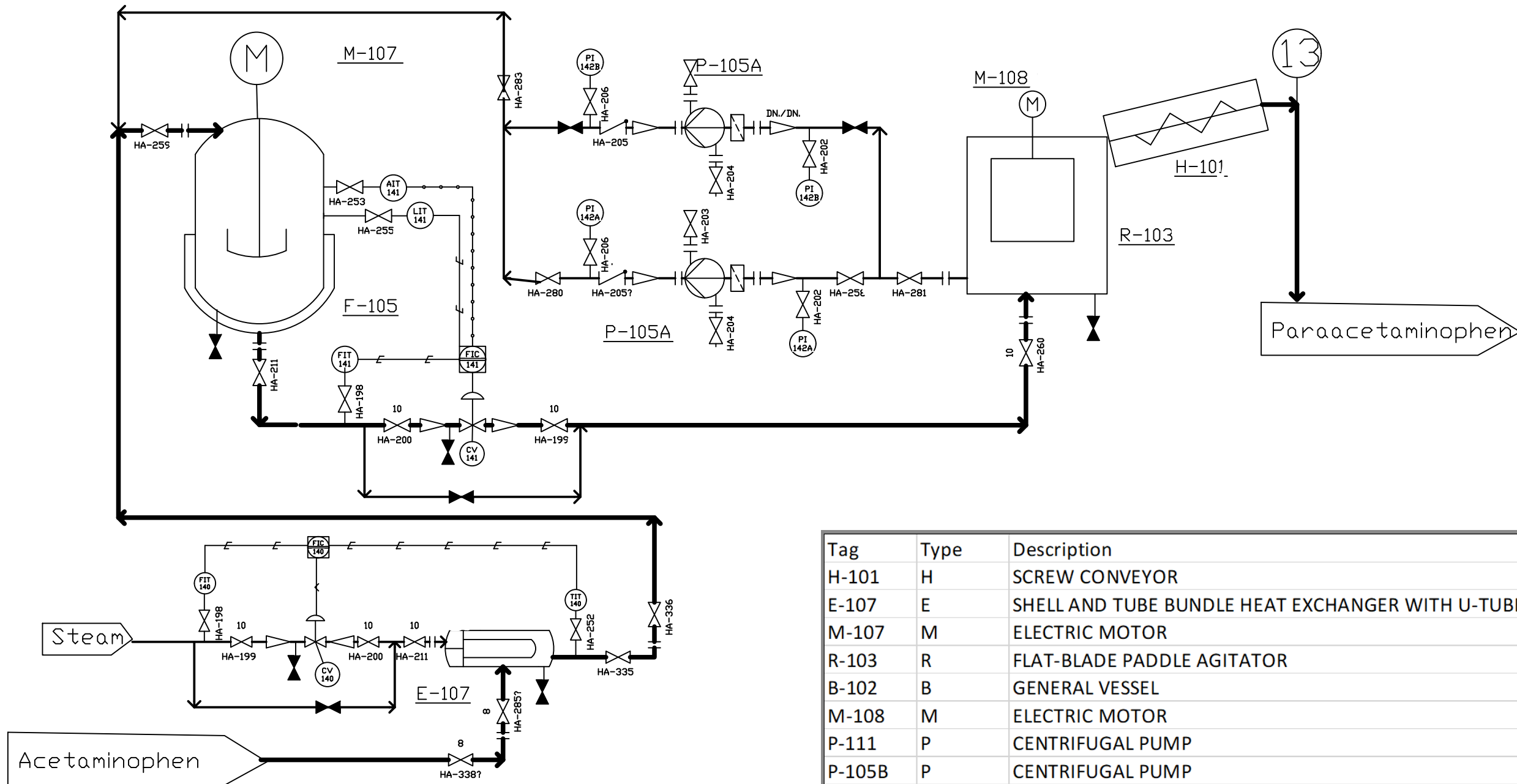




DESCRIPTION OF DIFFERENT STREAM LINES OF PIPING AND INSTRUMENTATION DIAGRAM				
Stream Name		10	11	14
Properties	Units			
Temperature	C	108	245.7279989	125.6578434
Pressure	bar	1	0.8	0.8
Mass Flows	kg/hr	7396.172	3165.844046	4230.327955
HYDROGEN				
	kg/hr	8.3755E-26	0	0
NITRO-BE				
	kg/hr	2.425514581	0.558008896	1.867505686
PAP				
	kg/hr	0.00028566	0.000285599	6.0728E-08
ANILINE				
	kg/hr	0.033373287	0.000206233	0.033167054
ACET-ANH				
	kg/hr	2962.907919	0.045040618	2962.862879
ACETIC				
	kg/hr	1259.74932	0.004395747	1259.744924
CETAMOL				
	kg/hr	3171.055588	3165.236109	5.819479062
WATER				
	kg/hr	7.06067E-08	3.41711E-15	7.06067E-08
SOLID				
	kg/hr	0	0	0
ETHANOL				
	kg/hr	0	0	0
Mass Density	gm/cc	0.969731039	0.75495794	0.002088845
Enthalpy Flow	cal/sec	-2300287.439	-420730.745	-1677725.733
Volume Flow	l/min	127.1172401	69.89007547	33753.33081
Mole Flows	kmol/hr	70.99750855	20.944	50.05350855

Tag	Type	Description
P-106B	P	CENTRIFUGAL PUMP
M-106	M	ELECTRIC MOTOR
E-108	E	SPRAY COOLER
V-101	V	GENERAL BLOWER
F-104	F	GENERAL GAS OR AIR FILTER
M-103	M	ELECTRIC MOTOR
W-103	W	BURNER FIRING SYSTEM
V-102	V	CONICAL HEADS VESSEL
P-107A	P	CENTRIFUGAL PUMP
E-109	E	SHELL AND TUBE BUNDLE HEAT EXCHANGER WITH U-TUBE
P-107B	P	CENTRIFUGAL PUMP
P-106A	P	CENTRIFUGAL PUMP
K-103	K	PLATE-TYPE HEAT EXCHANGER
P-115A	P	CENTRIFUGAL PUMP
E-112	E	SHELL AND TUBE BUNDLE HEAT EXCHANGER WITH U-TUBE
P-115B	P	CENTRIFUGAL PUMP

Manufacture of Paracetamol From Nitro Benzene				
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Tag	Type	Description
H-101	H	SCREW CONVEYOR
E-107	E	SHELL AND TUBE BUNDLE HEAT EXCHANGER WITH U-TUBE
M-107	M	ELECTRIC MOTOR
R-103	R	FLAT-BLADE PADDLE AGITATOR
B-102	B	GENERAL VESSEL
M-108	M	ELECTRIC MOTOR
P-111	P	CENTRIFUGAL PUMP
P-105B	P	CENTRIFUGAL PUMP
F-105	F	ANCHOR AGITATOR

Manufacture of Paracetamol From Nitro Benzene				
Kathmandu University	Sheet no	PREPARED BY	CHECKED BY	DATE
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